

— PRODUCT OVERVIEW

The Crypto Lending Platform That *Explains Itself*.

Every other platform is building faster pipes. Aetherum gives your members a lending decision they can actually understand — and that explanation is the moat.

12

DACS RISK
PILLARS

15

PATENTS PENDING

5,000+

CREDIT UNIONS
REACHABLE

SCHEDULE A DEMO

↓ DOWNLOAD PDF

— 01 • THE PLATFORM

Built for the member. *Deployed for the credit union.*

The market has been focused on speed for the institution. Aetherum is focused on trust for the member — and hands the credit union the infrastructure to deliver it.

LENDING

Crypto-Backed Lending

Members keep their crypto exposure, avoid a taxable event, and get USD liquidity — all through their trusted credit union. Aetherum handles the infrastructure. You own the relationship.

UNDERWRITING

Smart Review

AI-powered underwriting guidance evaluates the full borrowing position and explains it in plain language — recommended amount, risk level, and reasoning, instantly, at the point of decision.

PATENT PENDING

TAX INTELLIGENCE

Tax-Position Score & Optimizer (TPSOE™)

Scores the tax efficiency of each collateral position and ranks pledge assets to minimize member tax liability — a feature no D2C competitor offers institutions.

PATENT PENDING

— 02 · DACS

A risk assessment model built for crypto collateral.

DACS quantifies a borrower's position beyond traditional metrics — in language the member can actually understand. Twelve pillars of crypto risk assessment, backed by 15 patent-pending innovations, with a plain-language explanation behind every decision. Not a number on a screen; the reasoning behind it.

SEE THE 12 PILLARS →

02 · DACS — THE TWELVE PILLARS

1 Wallet Age
on-chain longevity

2 Diversification
portfolio spread

3 On-Chain Activity
active management

4 Volatility
collateral resilience

5 Tax-Position Score
TPSOE · patent pending

6 TAM Bonus
verified holdings

7 On-Chain Identity
CUMember · patent pending

8 Privacy Computation
FHE-DACS · patent pending

9 Dynamic LTV
DACS-LTV · patent pending

10 Ledger Collateral Control
patent pending

11 XLS-66 LoanBroker
patent pending

12 Key Governance & Audit
patent pending

Tier 1 members access up to 70% LTV, Tier 2 up to 65%, Tier 3 up to 55% — enforced at the blockchain layer, no manual override.

Compliance *built in*, not bolted on.

IDENTITY

On-Chain Member Identity

ERC-3643 + ERC-7943 compliant verification. Each member gets a portable ONCHAINID attestation — KYC verified, jurisdiction gated, collateral-scored on-chain. Built on the T-REX standard for regulated RWAs.

PATENT PENDING

PRIVACY

Privacy-Preserving Underwriting (FHE-DACS)

Encrypted risk assessment via homomorphic encryption — member financial data is never exposed in plaintext during underwriting.

PATENT PENDING

ENFORCEMENT

ERC-7943 Collateral Token

The official Ethereum Universal RWA Interface — on-chain eligibility checks, regulatory freeze, and forced transfer, all delegating through three ERC-3643 compliance modules.

PATENT PENDING

Every decision *examiner-ready* from day one.

RECEIPTS

On-Chain Compliance Receipts

SHA-256 hash of every decision written to Ethereum. Examiners receive a verification URL per decision; sensitive data stays off-chain. No other lending infrastructure provides this.

EXPLAINABILITY

DACS Explainability Layer

Auto-generated record after every scoring run — factor breakdowns, risk tier, peer comparison, plain-language rationale, and 12 CFR Part 723 citation. PDF export for NCUA submission.

PATENT PENDING

SUPERVISORY

Supervisory Control & Audit

Documented human-in-the-loop control over every position — NCUA-aligned event types, officer attribution, on-chain attestation, CSV examination export with Etherscan links.

PATENT PENDING

DILIGENCE

DDQ Engine

342-entry security knowledge base spanning NCUA guidance, custody, BSA/AML, and the full SIG framework. A CU IT team can complete a full vendor security review without a call.

LIVE

Money moves. *Collateral is watched.*

DISBURSEMENT

Multi-Rail Stablecoin

Pluggable multi-rail disbursement — USD & SGD across EVM networks, cross-chain transfer, and programmable MPC wallets. Migrating to Arc mainnet Q3 2026 for sub-second finality.

LIVE

SURVEILLANCE

Collateral Intelligence Engine (CIE™)

Every position monitored 24/7 — finality-aware on-chain event processing, live LTV, DACS updated on every price move, institution-authorized liquidation control.

PATENT PENDING

MONITORING

Native LTV Monitoring

BTC and ETH collateral polled every 60 seconds via Blockstream and Alchemy — no third-party dependency. 30-day rolling volatility feeds DACS directly.

LIVE

OVERSIGHT

Liquidation Escalation

Three-tier human oversight: Tier 1 auto-executes, Tier 2 pages your risk officer with a 30-min window, Tier 3 requires officer authorization for \$250K+ or 40%+ drops. Every action timestamped.

LIVE

A view *for every seat* at the credit union.

LENDING

Loan Officer Portal

Review applications, assess DACS scores, manage active loans, and action margin-call alerts — purpose-built for crypto-collateralized lending.

COMPLIANCE

Compliance Officer

Siloed KYC/AML queue, sanctions monitoring, and audit-ready reporting — full visibility and control without touching operational loan data.

REGULATORY

NCUA Examiner View

Read-only audit dashboard scoped per CU. Live positions, LTV ratios, liquidation events, full BSA/AML trail. Zero custody access, zero execution permissions.

MEMBERS

Member Crypto Insights

Consent-based visibility into the assets members already own — DACS eligibility scoring and an opt-in loan pipeline. See what's invisible today.

PATENT PENDING

Four steps. *You own the relationship.*

1

Member deposits crypto collateral into an institutional custody wallet — held directly by the credit union, never by Aetherum.

2

DACS scores the borrower and Smart Review AI evaluates the application against NCUA guidance — returning a structured decision with compliance notes and a full audit trail.

3

The credit union issues the loan using existing infrastructure. Aetherum handles the rest.

4

Automated monitoring manages margin calls, top-ups, and liquidation if needed — with human oversight on every large position.

GENIUS Act ready. Aetherum was built inside the NCUA's regulatory perimeter — on-chain identity, per-CU custody, and compliance-first infrastructure designed to operate within supervision, not around it.

08 · INTEGRATIONS & HOW WE'RE PAID

LIVE INTEGRATIONS

Jack Henry FIN

5,000+ CUs reachable

Corelation KeyStone

~290 CUs day one

Fiserv AppMarket

DNA platform

FIS Code Connect

IBS Open APIs

XRPL / XLS-66

RLUSD loan currency

Pharos · Circle CCTP

USDC, live testnet

REVENUE MODEL

- 01 Platform licensing fee (annual SaaS)
- 02 Per-loan origination fee
- 03 DACS API access for third-party integrations
- 04 RWA tokenization service fees (coming soon)

Credit unions keep 100% of loan interest income.

— 09 · GET IN TOUCH

Bring crypto-backed lending *to your credit union.*

Schedule a 30-minute demo.

We'll walk through the platform live — DACS, Smart Review, the audit-native trail — and answer any diligence questions.

[SCHEDULE A DEMO](#)

hello@aetherum.ai · aetherum.ai